The Effects of Cannabidiol Oil on Noninvasive Measures of Muscle Damage in Men

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ABSTRACT

COCHRANE-SNYMAN, K. C., C. CRUZ, J. MORALES, and M. COLES. The Effects of Cannabidiol Oil on Noninvasive Measures of Muscle Damage in Men. *Med. Sci. Sports Exerc.*, Vol. 53, No. 7, pp. 1460–1472, 2021. No previous study has investigated the applications of isolated cannabidiol (CBD) as a recovery aid in untrained human subjects after a bout of exercise-induced muscle damage. **Purpose:** This study aimed to investigate the effect of CBD oil on perceived muscle soreness, inflammation, and strength performance after eccentric exercise (ECC) of the elbow flexors. **Methods:** Thirteen untrained men (mean ± SD age, 21.85 ± 2.73 yr) performed 6 sets of 10 maximal ECC isokinetic muscle actions of the elbow flexors as part of a double-blind crossover design. Noninvasive (perceived soreness, arm circumference, hanging joint angle (JA), and peak torque (PT)) measures were taken before and after ECC, and 24, 48, and 72 h after ECC. All subjects completed both the supplement (CBD: 150 mg POST, 24 h, 48 h) and placebo (PLC: POST, 24 h, 48 h) condition separated by 2 wk. Four separate two-way repeated-measures ANOVA (condition [CBD vs PLC] × time [PRE vs POST vs 24 h vs 48 h vs 72 h]) were used to analyze perceived soreness, arm circumference, JA, and PT. One-way repeated-measures ANOVA were used to decompose significant interactions and main effects. **Results:** There was no condition–time interaction or main effect of condition ($P > 0.05$) for perceived soreness, arm circumference, JA, or PT. There were main effects for time for perceived soreness ($P = 0.000$, $\eta_p^2 = 0.71$) and JA ($P = 0.006$, $\eta_p^2 = 0.35$). **Conclusions:** The current dose of 150 mg CBD oil at POST, 24 h, and 48 h had no effect on noninvasive markers of muscle damage in the upper extremity. At the current dose and schedule, CBD oil may not be beneficial for untrained men as a recovery aid after exercise-induced muscle damage.
Key Words: CBD, EXERCISE, RECOVERY, SUPPLEMENT, PERFORMANCE

APPLIED SCIENCES

Cannabidiol (CBD) is one of the most studied components of the *Cannabis sativa* plant, but unlike Δ^9-tetrahydrocannabinol (THC), CBD is a nonintoxicating compound (1) that was removed from the banned substance list by the World Anti-doping agency in 2018. Although CBD has been a known derivative of *C. sativa* since 1940, it has recently gained popularity as a pharmaceutical intervention for epilepsy (2), arthritis, and neuroinflammation (3–5). In addition to its applications for clinical conditions, the interest in CBD has exponentially risen among athletes and nonathletes. For example, in a recent survey of 301 athletes' cannabis use and behaviors, 45% reported using CBD, with the majority presenting as novice (>3 yr) users (6). CBD products are now widely available in pharmacies, nutrition specialty, and retail stores despite limited data supporting their efficacy for nonclinical conditions, including markers of sports performance and injury recovery.

One common condition associated with a novel or high-volume exercise stimuli is delayed-onset muscle soreness (DOMS). DOMS is a sensation of discomfort often associated with inflammation that is typically felt within 24–72 h after bouts of unaccustomed exercise or high volumes of intense eccentric exercise (ECC) (7,8). Although some degree of muscle damage may be induced from any form of exercise, previous research has indicated that ECC is the most effective method of inducing damage, especially when applied to the elbow flexors (7,9,10). The associated symptoms of DOMS, such as soreness, muscle stiffness, aching pain, tenderness to palpation, and swelling, usually subside 5–7 d after exercise (8,11,12). One source of these symptoms is acute inflammation, which also been shown to occur as a response to the trauma seen in muscle after performing ECC (8,13–15). ECC has been shown to result in muscle damage, this acute inflammation, and DOMS (13,14). Thus, ECC is a well-accepted modality for investigating exercise-induced muscle damage (EIMD) and the efficacy of interventions to mitigate acute inflammation and DOMS associated with EIMD (9,16,17).

The most common way to measure inflammation is through muscle biopsies and blood draws. However, these techniques in themselves can impose further inflammation as they cause

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tissue damage through the insertion of the needle (18). Previous studies (9,18,19) have used noninvasive measures of muscle damage and inflammation such as visual pain scales (20,21), limb circumference, joint angles (JA) (19), and changes in peak torque (PT) to determine the degree of pain, swelling, and performance declines caused by ECC exercise. These measures, when paired with blood serum biomarkers, such as creatine kinase, the inflammatory interleukins (IL-1β, IL-6), or C-reactive protein have shown to be highly correlated and as such effective analogs for EIMD (9,20,21). Although invasive measures are the most popular method, noninvasive measures are suitable, are more affordable to measure in and outside of a laboratory setting, and do not impose further damage to tissues. Thus, the use of noninvasive measures of muscle damage does provide valuable information regarding the incidence of EIMD and DOMS and may be monitored for changes over time to determine the efficacy of a recovery intervention.

DOMS and associated inflammation have been shown to reduce athletic performance, which has fueled the development of preventable techniques in an attempt to attenuate these potential negative outcomes of unaccustomed exercise (22). Previous studies (23–26) utilizing treatments such as massage and nonsteroidal anti-inflammatory drugs (NSAIDs) for the treatment of EIMD or DOMS have produced equivocal results. Although the use of NSAIDs and massage are the most common treatments, each may vary in effectiveness in their ability to alleviate pain, soreness, and inflammation as a result of EIMD. Inflammation plays a key role in the muscle protein synthesis pathway, but it may also result in acute performance declines that are perceived negatively by most athletes and coaches (9,23,27). Thus, there is a need to investigate additional methods or supplement applications for the treatment of acute muscle damage after bouts of exercise.

CBD is becoming an increasingly popular medicinal treatment for pain and inflammation. CBD is said to work by binding to specific receptors (Vanilloid trpv1 and CB_2R) (3,28–30). For example, CBD has been shown to be a receptor agonist for the CB_2R receptor, which is located on cells associated with the immune system, cardiovascular system, and gastrointestinal tract, and plays a role in inflammatory immune responses. In addition, evidence is emerging that CBD interacts with the $5\text{-}HT_{1A}$ serotonin receptor, which is a target for antidepressant medications, and has some antioxidant properties (31). It has been used to treat various medical conditions such as arthritis, multiple sclerosis, and neuroinflammation (3–5). For example, Malfait et al. (5) found that oral administration of CBD was able to suppress the progression of arthritis because it presented immunosuppressive and anti-inflammatory properties. In addition, CBD has been shown to modulate inflammatory processes and affect markers commonly associated with EIMD (32). For example, CBD has been shown to reduce IL-1β, IL-6 as well as tumor necrosis factor (TNF-α), another biomarker for EIMD (33–35), which may have applications for neuromuscular inflammation related to DOMS. Although CBD has been shown to have anti-inflammatory and antihyperalgesic properties, there has been limited research on its effect after exercise (17,36).

The mechanisms underlying exercise-induced muscle soreness and inflammation have been studied for decades; however, limited studies have investigated the effects of a novel application for CBD and its effects on perceived muscle soreness and inflammation after exercise (17,36). To the author's knowledge, at the time of this study's undertaking, no previous study had investigated the effect of isolated CBD oil on EIMD in humans. Therefore, the purpose of this study was to investigate the effects of CBD on perceived muscle soreness, inflammation, and acute performance after bouts of ECC. In accordance with the previous anti-inflammatory applications for CBD oil, it was hypothesized that CBD may reduce perceived muscle soreness and inflammation, and enhance performance recovery.

METHODS

Participants. The study protocol was reviewed and approved by the University Institutional Review Board for human subject research before any recruitment or testing. Thirteen untrained men (mean ± SD age, 21 ± 2.73 yr, body mass, 78.63 ± 15.81 kg) volunteered to participate and completed both conditions of the study as part of a double-blind crossover design. *A priori* power analysis for a repeated-measures design indicated that a sample size of 13 would be sufficient to demonstrate a significant result with an α level of $P < 0.05$, power of 0.8, and a significant effect size of 0.2 or higher based on significant effect size estimates for changes in PT after EIMD (37). An untrained individual was defined as those who had not participated in upper body resistance training for 6 months before participating in the study. The exercise protocol for inducing DOMS and the possible risks involved were thoroughly explained to all subjects. Subjects completed an informed consent document and a health history questionnaire, and were screened for musculoskeletal injuries involving the wrists, elbow, and shoulder joint as well as for any acute infections before participating in the study. Subjects were also screened for supplement and medication use and were excluded from the study if they consumed any substances that could confound the effects of the CBD supplement. Subjects were also asked to avoid using any other therapeutic modality (massage, ice compression, NSAIDs, etc.) and upper-body exercise during the course of the study, as these could have affected the quality of data collected.

Individuals were excluded from participation if they 1) had current and/or history of any musculoskeletal diseases or conditions affecting the upper body; 2) had a body mass index exceeding 30 $kg \cdot m^{-2}$; 3) failed the urinary drug screening for traces of THC at initial screening or any point during the duration of the study (Narcotest THC (ID Pharma, Paris, France)); 4) were consuming a regular antioxidant supplement; 5) had contraindications, allergy, or sensitivity to the active ingredient in the study supplement under investigation; 6) had any self-reported significant illness or condition that could be

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expected to interfere with the study parameters or study conduct, or put the subject at significant risk (e.g., an abnormal ECG, bleeding disorder, high blood pressure, anemia, decreased blood clotting ability, heart disease, anxiety, history of syncope or dizziness, or history of myocardial infarction; 7) had acute conditions known to affect inflammation markers' levels (acute infection, were taking antibiotics, acute postsurgical states, or recent severe trauma); 8) had used creatine within 9 wk before screening; 9) had participated in any drug or medical trial within 30 d leading into study participation; 10) were unable to complete a 3-d food log; 11) were unable to understand English or Spanish instructions; and 12) were allergic to vegetable/canola oil (placebo).

Experimental design. This study was a double-blinded, placebo-controlled, crossover design, and took place in the University's Human Performance Lab (Fig. 1). All experimental sessions occurred at the same time of day ± 30 min. On visit 1, subjects were randomly assigned to one of the two experimental groups, 150 mg (CBD) or placebo (PLC), based on their assigned subject number and corresponding randomization code. Subjects were familiarized with the supplement administration instructions before any testing or measurements. Subjects were also instructed on how to complete a 3-d food recall log to be analyzed using myfitnesspal.com calorie analysis and were familiarized with the isokinetic dynamometer (Biodex, Corp., Shirley NY) and the Borg 6–20 RPE scale using standardized instructions (38). During the dynamometer familiarization, subjects were asked to give different levels of effort (0%–100% maximal voluntary isometric contraction (MVIC)) and to assess their RPE to help establish anchoring cues for the RPE scale (38). The subjects' heights (in meters) were measured using a stadiometer (Novel Products Inc., Rockton, IL), and weight and fat percentages were measured via bioelectrical impedance spectrometry (Tanita Corp.,

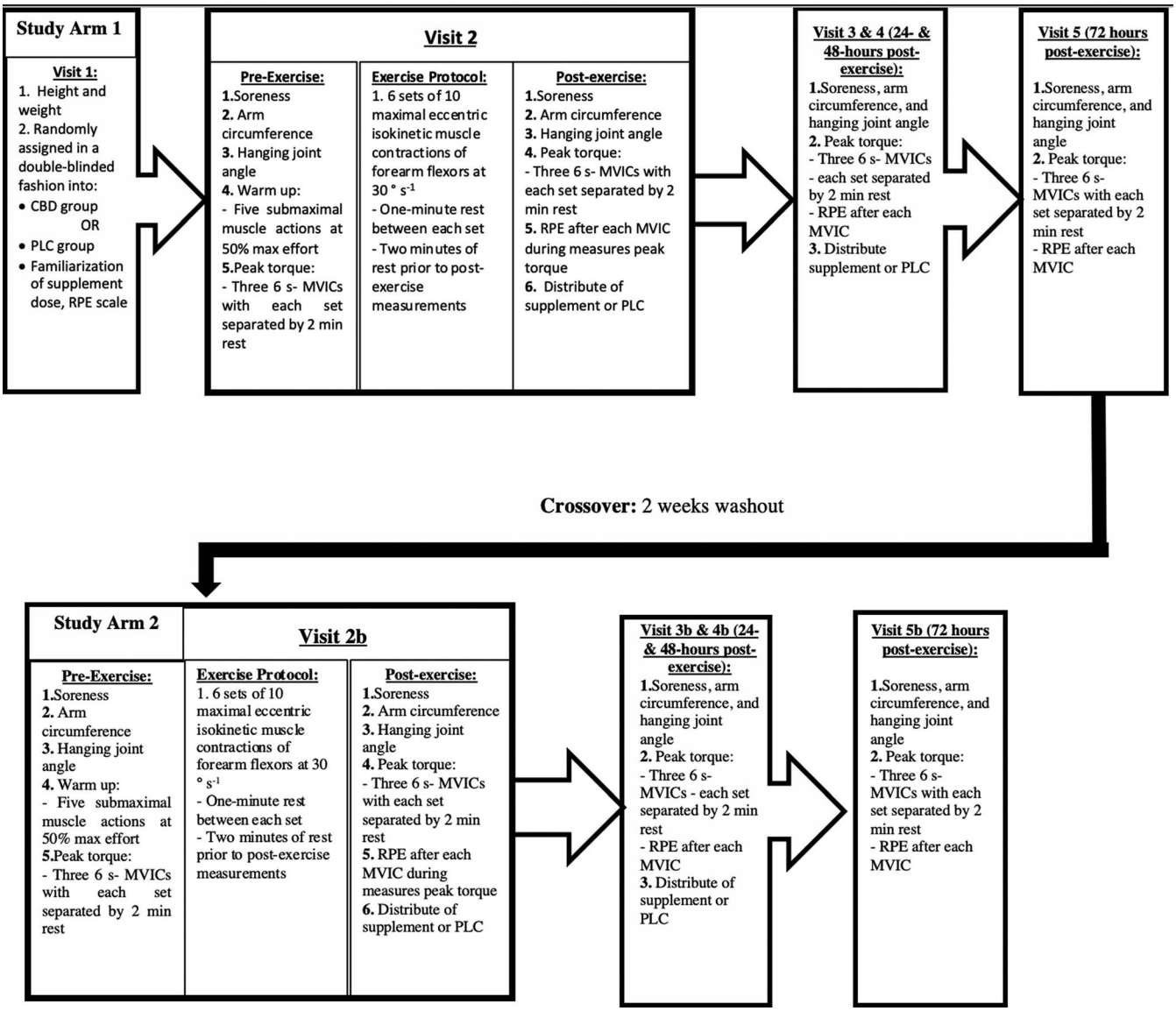

FIGURE 1—Study design.

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Tokyo, Japan). On all experimental visits, subjects completed a THC urinary screening test (Narcotest THC (ID Pharma)). Upon completion of the aforementioned test, the following variables were measured during the first experimental visit for each condition (CBD, PLC): resting blood pressure, perceived soreness using a visual analog scale, arm circumference of the randomized test limb (counterbalanced across conditions) using a Gulick tape measure (Mabis Healthcare, Waukegan, IL), hanging JA using a goniometer (Smith and Nephew Rolyan Inc., Menomonee Falls, WI), and PT using a calibrated isokinetic dynamometer (Biodex, Corp.). Subjects then performed a warm-up of five submaximal muscle actions involving the elbow flexors at 50% of maximal effort before conducting measurement of MVIC PT. All measurements (perceived soreness, JA, arm circumference, RPE, and PT) were taken immediately before ECC (PRE), 2 min after ECC (POST), and at 24 (visit 3), 48 (visit 4), and 72 h (visit 5) after muscle-damaging exercise protocol. The ECC protocol occurred only on the first visit of each testing cycle.

After POST measures, participants were given their blinded capsule canisters containing either experimental supplement (non-THC CBD) or PLC (vegetable oil). The primary investigator and all researchers involved were blinded to the contents of the canisters. Randomization and blinding code were created and maintained by an outside agent. The experimental group CBD consumed two doses of 75 mg (three capsules per dose; 150 mg total) of CBD orally per day separated by 8 h. Participants were instructed to take their first dose within 1 h after completing visit 2, and subsequent dosages for CBD or PLC were consumed 8 ± 1 h after the initial dosage on visit 2, visit 3, and visit 4. The PLC groups received an identical dosage (three capsules per dose) and administration of a capsule filled with non-CBD containing vegetable oil. Vegetable oil, without traces of flaxseed n-3 oil was selected as the PLC because it contained no known antioxidants or other properties that may affect anti-inflammatory pathways. Although flaxseed-rich vegetable oil has previously been shown to affect markers of inflammation (39), specifically those with high levels of n-3 fatty acid derivatives, the vegetable oil selected as the placebo was sunflower and soybean based with no effects on inflammatory pathways in the quantities administered. On visit 5, participants returned to the laboratory and completed all POST ECC protocol measures but did not receive another dose of the supplement or placebo. After a washout period of 2 wk, participants returned to undergo either the CBD or PLC condition, whichever was not administered during the initial visits. During the crossover, the subject performed the same exercise protocol and series of measurements as in the initial visits, but all exercises and measures were performed on the contralateral arm.

Supplementation. CBD capsules: CBD (150 mg) were divided into two separate 75-mg doses to be separated by 8 h. Participants consumed three 25-mg CBD gel capsules (Green Roads, Davie, FL) for a total dose of 75 mg per 8 h. Previous research has shown efficacy for low doses of 1.5 $mg \cdot kg^{-1}$ body mass up to 10 $mg \cdot kg^{-1}$ (40) of CBD for its effects on markers of inflammation. The dosage in the current study was selected to be within this range (2.0 $mg \cdot kg^{-1}$ average), as limited information on the use of this supplement for this purpose in human subjects is available. The time course for time to maximum concentration has been reported to be between 1.5 and 3 h for CBD dosages of 150–300 mg, and terminal-half life may last as long as 17 h after supplementation (41,42). Therefore, the absolute dose administered after each visit (150 mg) fell within this range and was expected to present in circulation up until the next testing and supplement administration session (41,42). The hemp-derived CBD isolate used in the present study was independently shown (Kaycha Labs, Davie, FL) to have 4.097% CBD and 4.108% total cannabinoids (0% THC) in the sample tested with 25 mg of CBD per capsule. Placebo capsules were gelatin capsules filled with 25 mg of 100% certified organic vegetable oil (Spectrum Naturals, Lake Success, NY) and allocated in the same quantities (six capsules per day) as the CBD condition. Placebo capsules looked identical in appearance to the CBD capsules. Vegetable oil was selected as a safe PLC, which would not significantly affect inflammation in the quantities provided (75 mg per dose), yet appear identical in appearance and texture to the active, CBD capsules (43). All supplement and placebo capsules were distributed in blinded containers with predetermined coding set by the independent party responsible for controlling the blind. Neither researcher nor participant knew the contents of each capsule canister. To facilitate allocation concealment, the independent party who formulated the blind maintained the blind record and black canisters, which hid the contents from the experimenters during assignment. All capsules and canisters were identical in appearance. Each black canister, filled with either CBD or PLC, as designated by the blinder, was given a 1 or 2 designation, and each subject number was randomly assigned 1 or 2 as their starting condition. Subject numbers were assigned as subjects were enrolled, and therefore, there was no bias related to allocation of condition assignment. Each participant only received the capsules associated with each visit; therefore, three separate canisters were distributed to each participant during condition 1 and condition 2 of the study. The participants did not take any of the study product or placebo during the washout period. As the terminal half-life for 250–300 mg of CBD has been shown to be 14–17 h (41,42), the 2-wk washout was deemed sufficient to remove any trace of active supplement before completing the crossover. Compliance was assessed by having each participant log the timing of their dosages and returning the study canister at each subsequent visit. Participants who failed to consume 80% of the study drug within the 16-h supplementation window was considered not in compliance. In addition, THC urinary screenings were completed during all experimental visits to ensure participants were not consuming substances, which may confound the study results.

ECC protocol. To induce soreness, subjects performed 6 sets of 10 maximal ECC isokinetic muscle contractions of the elbow flexors at $30° \cdot s^{-1}$ using a Biodex dynamometer

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TABLE 1. Descriptive characteristics of the subjects (n = 13).

Variable	Mean ± SD	Range
Age, yr	21.9 ± 2.7	19.0–26.0
Height, cm	173.3 ± 9.0	163.8–186.7
Weight, kg	78.7 ± 15.8	47.6–109.3
Body fat, %	21.3 ± 5.3	12.0–30.0
BMI, $kg \cdot m^{-2}$	26.2 ± 3.0	31.3–60.2
CBD, $mg \cdot kg^{-1}$	2.0 ± 0.5	1.4–3.2

BMI, body mass index.

(Biodex, Corp.). Subjects performed the first round of experimental exercise visits using their dominant arm (determined by handedness) and nondominant arm during the crossover portion of the study. The exercise was performed with the hand in a neutral position, and subjects were placed in a sitting position on the Biodex dynamometer with the fulcrum of the Biodex lever arm aligned on the lateral size of the subject's elbow. Subjects began each eccentric muscle action at an angle of 50° at the elbow and were instructed to give a maximal effort while resisting the lever arm until they reached an extended angle of just under 180°. Subjects were then assisted through the concentric portion of the movement range to ensure significant energy was not expended to return to the 50° starting position. To ensure maximal effort, the same tester gave verbal encouragement and subjects could view their torque production using the Biodex system monitor. One minute of rest was given between each set, and before measurements that were taken after exercise, 2 min of rest was given (9).

Instrumentation. Muscle soreness was measured using a 10-cm visual analog scale adopted from Bobbert et al. (44), which included verbal descriptions of pain. To indicate the level of soreness felt when the elbow and forearm were extended, the subject marked the scale corresponding to their pain. Perceived soreness was then measured in centimeters using a ruler across the scale. Arm circumference was measured at the midbelly of the bicep. The measurement was taken with the arm horizontally abducted and the forearm extended (9). The arm utilized was randomized across both groups, with the starting arm determined by handedness based on throwing preference. Hanging JA between the forearm and arm was measured using a goniometer (Smith and Nephew Rolyan Inc., Menomonee Falls, WI). For each measurement, the axis of rotation of the elbow joint was aligned with the axis of the goniometer. The proximal arm of the goniometer was aligned with the acromion process of the scapula, and the distal arm was aligned with the styloid process of the ulna (9). MVIC PT was measured on a Biodex dynamometer (Biodex, Corp.). The Biodex has been shown to be reliable and valid for the measurement of MVIC in both trained and untrained populations (45,46). A random subsample of untrained subjects (n = 8) was used to calculate the coefficient of variation for MVIC rep 1 versus MVIC rep 2 and was found to have ranged from 0% to 8.0%, with most subjects ranging between 0% and 3.0%, which represents a good coefficient of variation precision of measurement value. Subjects performed a warm-up of five submaximal muscle actions at 50% of maximal effort before conducting measurement of MVIC PT. After warming up, subjects performed three, 6-s MVICs of the elbow flexors seated, with the hand in a neutral position. As with the methods of Jenkins et al. (9), the angle between the arm and forearm was set at 115° and tension was released from the lever arm before initiation of MVICs. Subjects were instructed to give maximal efforts and contract their arm as hard and fast as possible. Each set was separated by 2 min of rest. The highest torque measurement out of the 3 was used for analysis (9). RPE was measured immediately after each 6-s MVIC, as RPE may be used as an indicator of maximal effort (38). The same investigator measured each dependent variable for each subject to maintain intersubject reliability.

Statistical analyses. Five separate two-way repeated-measures ANOVA tests (condition [CBD vs PLC] × time [PRE vs POST vs 24 h vs 48 h vs 72 h]) were used to analyze perceived soreness, JA, arm circumference, RPE, and PT. Follow-up one-way repeated-measures ANOVA was used to decompose significant interactions and main effects for time, whereas one-way between-factor ANOVA was used for condition. Significant one-way ANOVA was followed by pairwise comparisons using Sidak–Bonferonni error correction for multiple comparisons. Partial η squared (η_p^2) and Cohen's d effect sizes were calculated for each ANOVA and pairwise comparison, respectively, and presented for all significant results. The Mauchly sphericity test was used to test assumptions of homogeneity of variance. If this was violated, the Greenhouse–Geisser value was used to adjust degrees of freedom to increase the critical value of the F ratio. Normality was assessed using histogram plots with a normal distribution curve of best fit and inspection of QQ residual plots. All statistical analyses were completed using SPSS (Version 25; IBM, Armonk, NY) using an *a priori* α level of 0.05 to determine the threshold for significance.

RESULTS

Subject descriptive characteristics can be found in Table 1. For perceived soreness, there were no significant interactions between the experimental factors (condition × time; $F(4,48) = 0.053$,

TABLE 2. Results of repeated-measures ANOVA (n = 13).

Variable	P	F	η_p^2
Perceived soreness			
Drug	0.954	0.003	0.000
Time	>0.00	29.49	0.711
Drug × time	0.913	0.053	0.004
Arm circumference			
Drug	0.987	0.000	0.000
Time	0.134	2.157	0.152
Drug × time	0.418	0.998	0.077
JA			
Drug	0.322	1.068	0.082
Time	0.006	6.443	0.349
Drug × time	0.918	0.234	0.019
PT			
Drug	0.533	0.412	0.033
Time	0.180	1.869	0.135
Drug × time	0.951	0.055	0.005
RPE			
Drug	0.504	0.474	0.038
Time	0.004	4.374	0.267
Drug × time	0.760	0.466	0.037

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$P > 0.05$) or significant main effect for condition ($F(1,12) = 0.003$, $P > 0.05$; Table 2). However, there was a main effect for time ($F(4,48) = 29.487$, $P = 0.000$, $\eta_p^2 = 0.71$). *Post hoc* pairwise comparison revealed that perceived soreness measures were significantly less at PRE (0.04 ± 0.24 cm) compared with POST, 24, 48, and 72 h posttest (4.07 ± 0.61, $d = 3.585$; 4.97 ± 0.48, $d = 4.908$; 4.79 ± 0.39, $d = 5.225$; 3.94 ± 0.57 cm, $d = 3.57$; $P = 0.000$; Fig. 2).

For arm circumference, baseline arm circumference for CBD condition was 31.8 ± 2.8 cm and PLC was 31.4 ± 2.6 cm ($P > 0.35$). There were no significant interactions between the experimental factors (condition × time; $F(4,48) = 0.998$, $P > 0.05$), significant main effect for condition ($F(1,12) = 0.00$, $P > 0.05$) or significant main effect for time ($F(4,48) = 2.157$, $P > 0.05$; Table 2, Fig. 3). For JA, there were no significant interactions between the experimental factors (condition × time; $F(4,48) = 0.234$, $P > 0.05$) or significant main effect for condition ($F(1,12) = 1.068$, $P > 0.05$; Table 2). However, there were main effects for time ($F(4,48) = 6.443$, $P = 0.006$, $\eta_p^2 = 0.35$). The *post hoc* pairwise comparison revealed that PRE measures were significantly greater (165.42° ± 1.52°) compared with POST and 24 h posttest, respectively (160.12° ± 2.25° and 158.92° ± 2.51°; ($P < 0.05$, $d = 0.379$; $P = 0.006$, $d = 0.37$; Fig. 4).

For PT, there were no significant interactions between the experimental factors (condition × time; $F(4,48) = 0.055$, $P > 0.05$) or significant main effect for condition ($F(1,12) = 0.412$, $P > 0.05$; Table 2). There was no main effect for time ($F(4,48) = 1.869$, $P > 0.05$; Fig. 5). For RPE, there were no significant interactions between the experimental factors (condition–time; $F(4,48) = 0.466$, $P > 0.05$) or significant main effect for condition ($F(1,12) = 0.474$, $P > 0.05$; Table 2). However, there was a significant main effect for time ($F(4,48) = 4.374$, $P = 0.004$, $\eta_P^2 = 0.27$). *Post hoc* pairwise

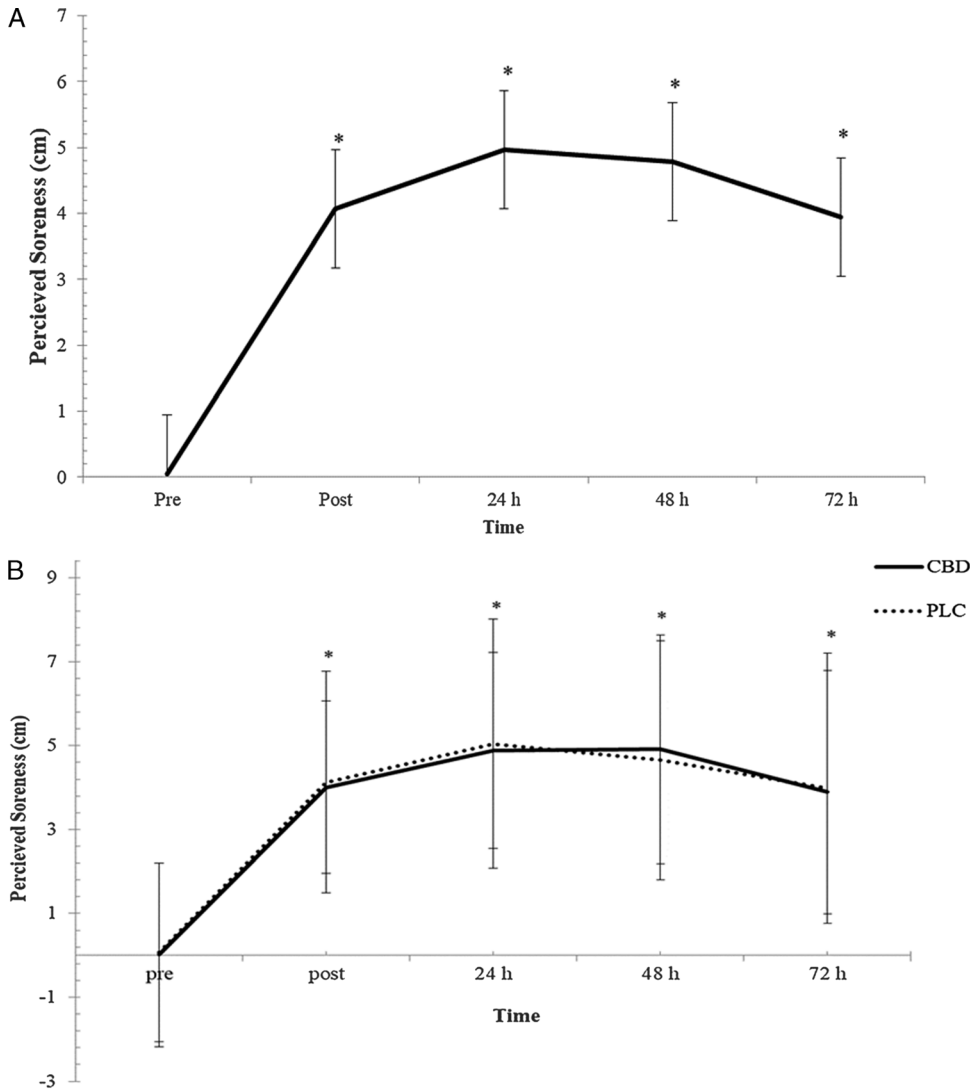

FIGURE 2—Recovery of perceived soreness. A, Data presented are marginal means for time for perceived soreness in both the CBD and PLC groups assessed before and after exercise and 24, 48, 72 h after exercise. B, Data presented are means ± SD of the mean for perceived soreness in the supplement (*solid line;* CBD) and placebo (*dotted line;* PLC) groups assessed before and after exercise and 24, 48, 72 h after exercise. *Denotes a value that was significantly greater than pretest for the main effect of time ($P < 0.05$).

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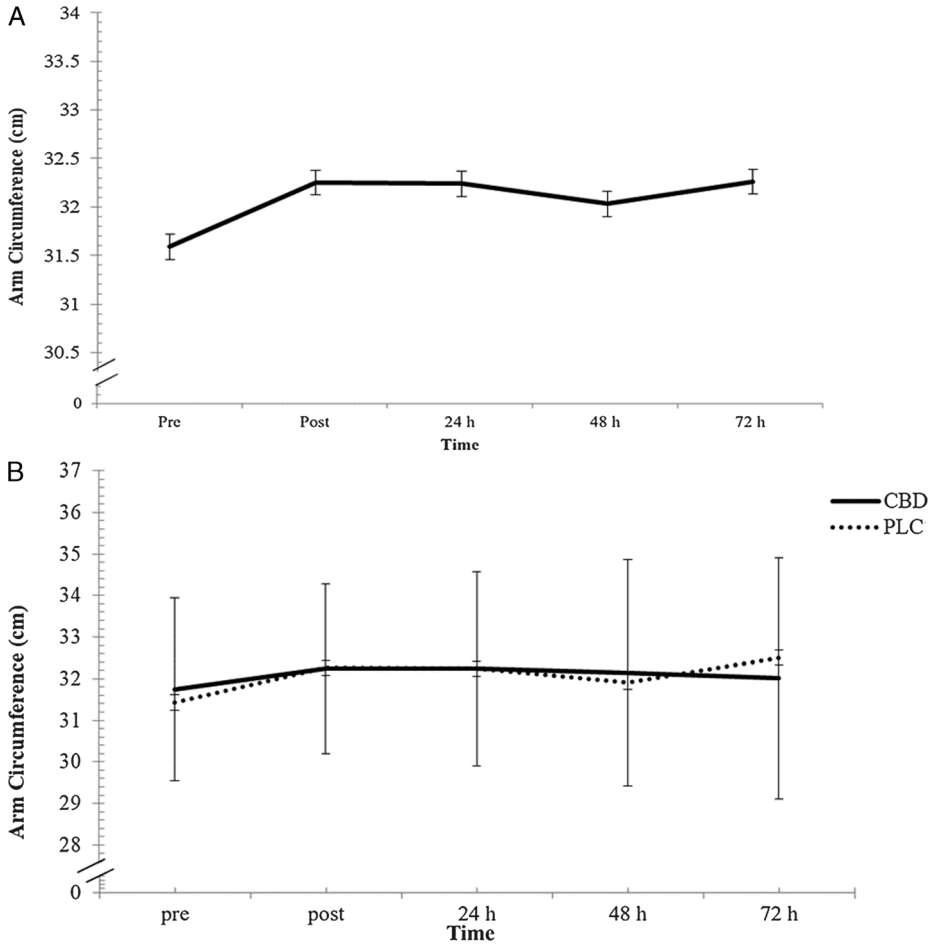

FIGURE 3—Recovery of arm circumference. A, Data presented are the marginal means for time for arm circumference (in centimeters) for both the CBD and PLC groups assessed before and after exercise and 24, 48, 72 h after exercise. B, Data presented are means ± SD of the mean for arm circumference (in centimeters) in the supplement (*solid line;* CBD) and placebo (*dotted line;* PLC) groups assessed before and after exercise and 24, 48, 72 h after exercise.

comparison revealed that when collapsed across condition, RPE was significantly less (P = 0.027, D = 2.97) at PRE (14.77 ± 0.54) compared with measures at 72 h posttest (16.04 ± 0.66; Fig. 6).

DISCUSSION

The results of the present study did not support the hypothesis that CBD oil supplementation would have an effect on noninvasive markers of muscle damage and inflammation after an ECC protocol. Within the present study, the changes observed across time were consistent with patterns expected as a result of ECC-induced muscle damage. Previous studies (9,13,16,47–49) have indicated that performing repetitive ECC exercise results in muscle damage in the working muscle, leading to muscle soreness and inflammation. EIMD results from mechanical injury and biochemical mechanisms. Damage to structural components of sarcomeres, such as the z-line and contractile filaments, may cause the release of proteolytic enzymes and proinflammatory cytokines associated with an acute inflammatory response (50). Thus, performing high-volume ECC exercise provides the modality to investigate EIMD and associated noninvasive markers of that damage.

The exercise protocol used in this study was similar to that of previous studies (9,16,51) used to induce muscle damage in the forearm flexors. In the present study, there was a 12% reduction in PT when collapsed across conditions from pretest to posttest. Although statistical significance was not met at the 0.05 level across all visits, absolute PT remained depressed at 24, 48, and 72 h post across both conditions (Fig. 5). Reductions in PT were, however, less than those reported in previous studies (9,16). For example, Beck et al. (16) reported between 21% and 43% reductions in PT, whereas Jenkins et al. (9) reported PT reductions between 23% and 44% after ECC exercise. The present study utilized untrained participants, which was similar to Jenkins et al. (9). It has been reported (50) that untrained participants may require verbal encouragement during maximal voluntary tasks, such as a MVIC, to ensure that accurate measures of maximal voluntary PT are achieved before the test. For example, the authors (50) found a 5% increase in peak force when untrained individuals were given verbal encouragement during isometric contractions of the elbow flexors. Verbal encouragement was provided that was

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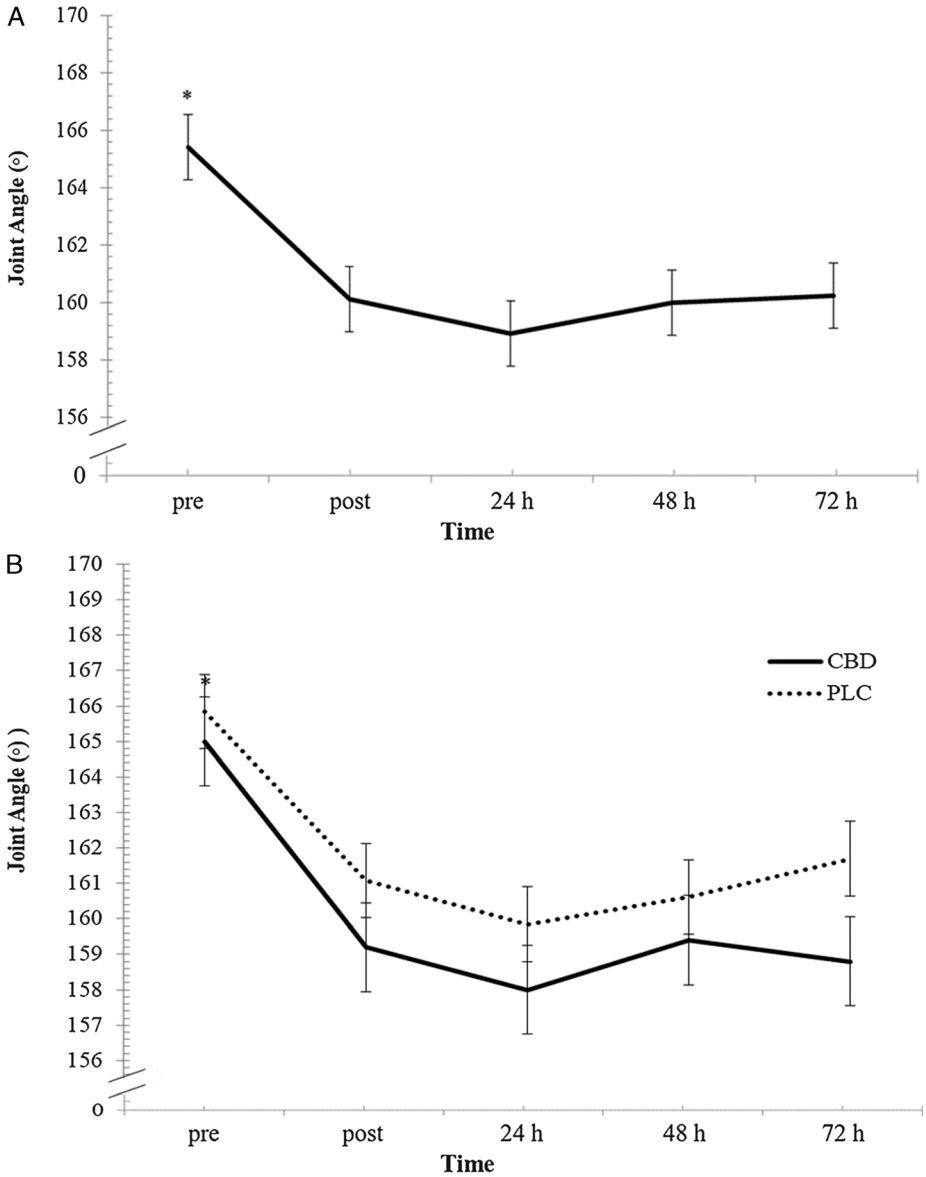

FIGURE 4—Recovery for JA. A, Data presented are the marginal means for time for JA (in degrees) for both CBD and PLC groups assessed before and after exercise and 24, 48, 72 h after exercise. B, Data presented are means ± SD of the mean for JA (in degrees) in the supplement (*solid line;* CBD) and placebo (*dotted line;* PLC) groups assessed before and after exercise and 24, 48, 72 h after exercise. *Denotes a value that was significantly greater than posttest, and 24 h posttest for the main effect of time ($P < 0.05$).

consistent with current recommendations and in agreement with the methods of Beck et al. (16) and Jenkins et al. (9). In addition, pre-MVIC measures were found to be consistent across conditions. Despite differences in the magnitude of change reported for PT, time-dependent changes in PT after high-volume ECC exercise has been shown to be a potent indicator of muscle damage (52). Thus, the change in PT, although limited, indicates that muscle damage was induced as a result of the ECC exercise protocol.

Acute muscle soreness occurs during or immediately after performing high-volume eccentric or novel exercise. It has been demonstrated (11,50) that acute soreness may progress into DOMS within 24 h before subsiding ~5 or fewer days after exercise. Pain, stiffness, tenderness, and swelling are all common symptoms of exercise-induced muscle soreness (12). Stiffness after exercise is believed to occur as a result of connective tissue damage, which often accompanies EIMD. Damage to connective tissues increases mechanical sensitivity within nociceptors and stretch receptors in the muscle resulting in discomfort and pain upon activation (50). The present study demonstrated increases in perceived soreness from pre-ECC to post-ECC and at 24, 48, and 72 h after completion of the ECC protocol. In addition, peak soreness occurred within 24–48 h after performing the exercise protocol, which coincided with a decrease in JA from pretest to posttest and 48 h posttest (Figs. 2, 4) and was consistent with the time course of DOMs. In a similar study investigating the effectiveness of a tobacco-derived supplement on muscle damage,

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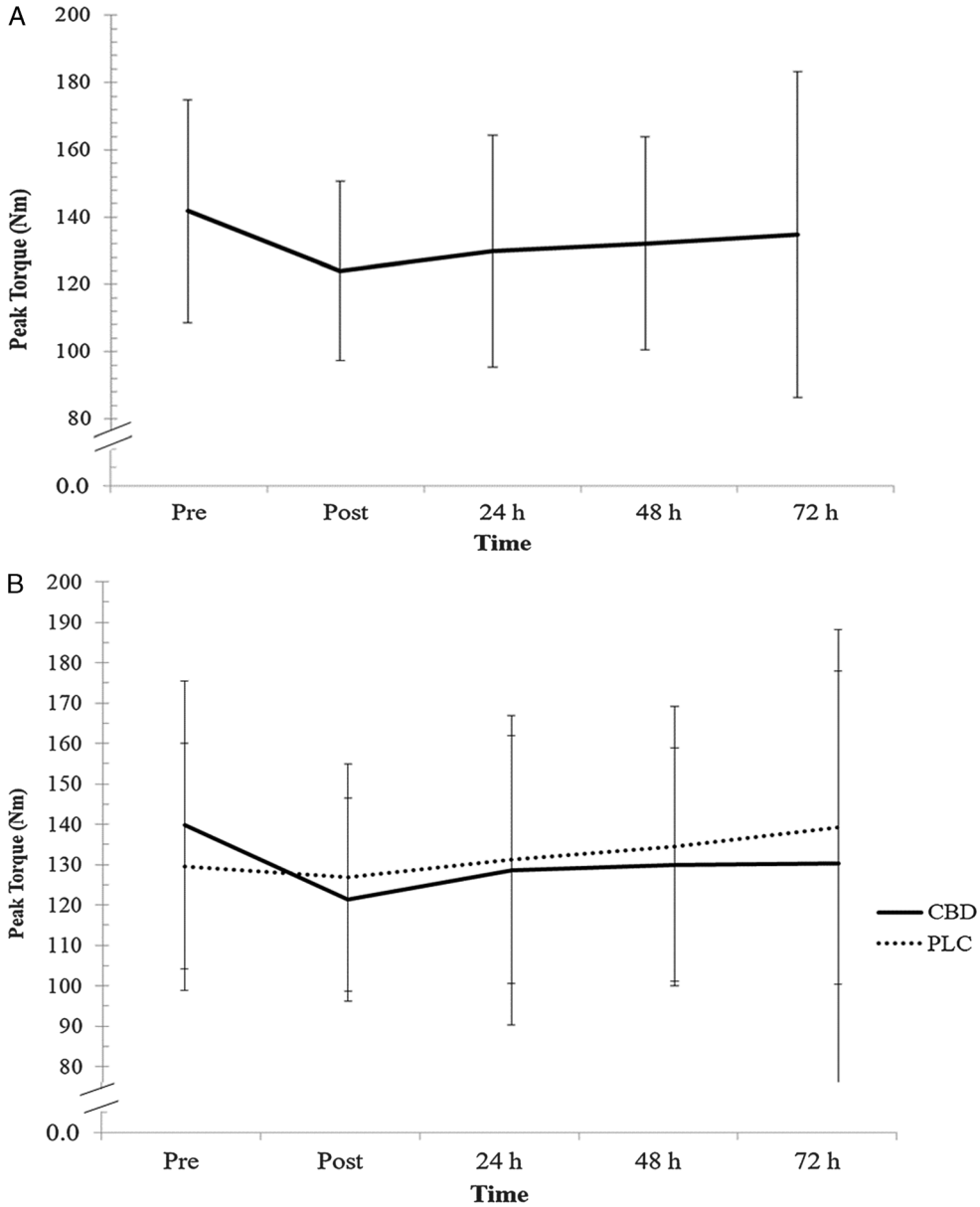

FIGURE 5—Recovery for PT. A, Data presented are marginal means for time for PT in both the CBD and PLC groups assessed before and after exercise and 24, 48, 72 h after exercise. B, Data presented are means ± SD of the mean for PT (newton-meters) in the supplement (*solid line;* CBD) and placebo (*dotted line;* PLC) groups assessed before and after exercise and 24, 48, 72 h after exercise.

Jenkins et al. (9) reported increases in soreness and decreases in JA after the same ECC procedures but no effect of supplementation on these time-dependent changes. These findings were similar to the present study in that changes in perceived soreness and JA did not differ between CBD and PLC conditions. Previous research (53) has demonstrated that CBD interacts with adenosine A2A receptors, which can protect tissue from inflammatory damage. This effect, at doses as low as 1 $mg \cdot kg^{-1}$, was most markedly shown via downregulation of TNF-α (53). It has also been reported (54) that CBD's interaction with the CB2 cannabinoid receptor can disrupt the arachidonic acid inflammatory pathway involved in the development of inflammation and edema. The bulk of this direct, inflammatory evidence comes from rodent models (53,54). Thus, there may be a dose-dependent difference in the effectiveness of CBD on A2A and CB2 receptors in response to the persistent inflammation associated with EIMD. In the present study, the lack of condition-specific differences in responses for perceived soreness and JA supported the manifestation of acute muscle damage, but not the efficacy of the current dose and schedule of CBD oil as a means of reducing these noninvasive measures of muscle damage.

Arm circumference did not change as a function of drug condition, which indicates that any active ingredient in CBD did not affect swelling more than the control. There was, however, a transient increase in arm circumference from pretest to posttest (Fig. 3) that tracked the modest decline in PT (Fig. 5). It is common to observe increases in limb circumference

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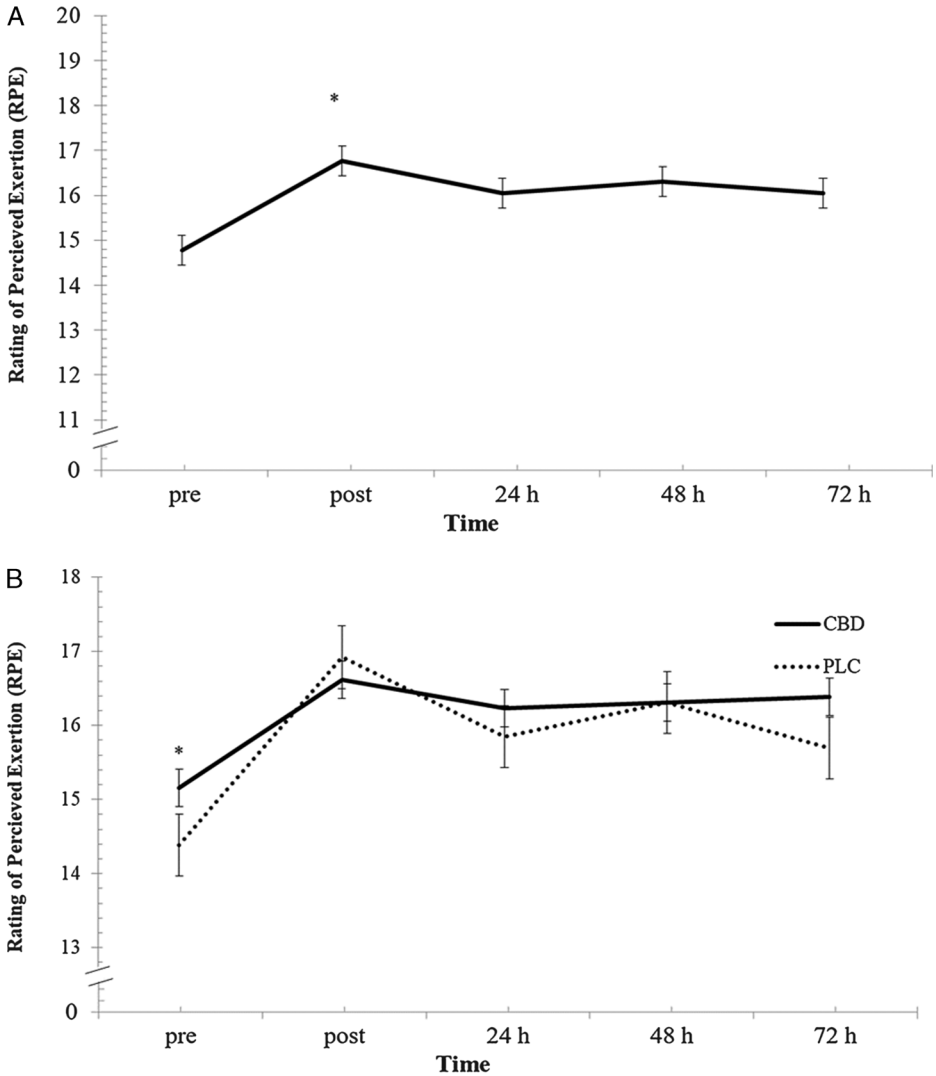

FIGURE 6—Recovery for RPE. A, Data presented are marginal means for RPE for condition groups (CBD and PLC) assessed before and after exercise and 24, 48, 72 h after exercise. B, Data presented are means ± SD of the mean for RPE in the supplement (*solid line;* CBD) and placebo (*dotted line;* PLC) groups assessed before and after exercise and 24, 48, 72 h after exercise. *Denotes a value that was significantly less than 72 h posttest for the main effect of time ($P < 0.05$).

following acute exercise because of cell and fluid migration from circulation into the interstitial spaces that surround muscle fibers (5). Any fluid shift that may have resulted from ECC exercise and subsequent arm swelling (arm circumference) detected immediately after ECC protocol was not deemed to be statistically significant at 24, 48, or 72 h post. It has been shown in rodent models (29) that CBD oil can exert an anti-inflammatory effect reducing edema in response to oral dosages of 5–40 $mg \cdot kg^{-1}$ for a sequence of 72 h after induced inflammation via carrageenan injection. These effects were attributed to the interaction of CBD with A2A receptors and a reduction of cyclooxygenase activity. In the present study, the average dose received per kilogram body mass was approximately 2.0 $mg \cdot kg^{-1}$, which was less than the average doses shown to be effective in rodent models but similar to doses shown to be effective at reducing neuroinflammation in human subjects (41). These findings indicate that there may be more variation in effective dose requirements across different forms of inflammation in human models (neuro vs EIMD) and those models of inflammation tested in rodent models. In addition, the results of the present study indicate that the transient increase in arm circumference reflected a transient fluid shift as a result of local swelling and increases in intramuscular pressure (55) common to acute ECC exercise, but sit was likely not dependent on the PLC effect or CBD oil supplementation in the current dose and administration schedule.

In recent years, the application of CBD for the treatment of various medical conditions has been on the rise. Previous studies (4,27) have reported improvements in symptoms of arthritis, multiple sclerosis, and neuroinflammation with the use of CBD oil. For example, administering 5 $mg \cdot kg^{-1}$ of CBD per day via i.p. or 25 $mg \cdot kg^{-1}$ given orally was shown to have an optimal effect in the suppression of gene-related and inflammatory markers associated with arthritis in mice, specifically glial fibrillary acidic protein, inducible nitric oxide

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synthase (iNOS), and IL-1β (4). CBD has also been shown to reduce Aβ-induced neuroinflammation after administering 2.5 or 10 mg·kg^{-1} via i.p. by impairing iNOS and IL-1β protein expression. β-amyloid is found within complex extracellular lesions of senile plaques, a hallmark of Alzheimer's disease. Activation of Aβ-induced glial cells triggers an inflammatory response, in which there is a release of neurotoxic cytokines (e.g., IL-1β). Interleukin-Ib plays a significant role in the cytokine cycle of both cellular and molecular events that are accountable for neurodegenerative consequences, such as the synthesis and processing of amyloid precursor proteins, and the activation of astrocytes with a consequent overexpression of iNOS and an overproduction of nitric oxide. Nitric oxide is a free radical that is short-lived and diffusible and supports the detrimental progression of Alzheimer's disease (56). In addition, Costa et al. (56) reported that pain within an acute carrageenan-induced inflamed rat paw was significantly reduced with oral doses of CBD as low as 5 and 7.5 mg·kg^{-1}. The authors (56) also reported that edema within the injected rat paw decreased with dosages of 5, 7.5, 10, 20, and 40 mg·kg^{-1} of CBD. Although this study did not investigate a neurogenerative disease condition, these known markers targeted by CBD may provide insight into their mechanism of action during other inflammatory conditions, such as EIMD, which is characterized by swelling, inflammation, and mild edema in humans.

In the present study, CBD oil was administered after a bout of EIMD in an attempt to ameliorate soreness, inflammation, and performance declines typical after bouts of repeated, muscle damaging, ECC exercise. After performing 6 sets of 10 repetitions of ECC muscle actions of the forearm flexors, acute decreases in performance, JA, and increases in swelling and RPE were observed. However, the results of this study indicated that a dose of 150 mg (or approximately 2 mg·kg^{-1}) of non-THC containing CBD oil administered three times from POST to 48 h post (total dose of approximately 6 mg·kg^{-1}) had no significant effect on noninvasive measures of muscle damage and inflammation. For example, the present study found no significant condition–time interactions or main effects for condition for perceived soreness, arm circumference, JA, PT, and RPE ($P > 0.05$) The lack of significant interactions or main effects for condition in the present study indicated that there was no placebo or drug effect related to the ingestion or perceived ingestion of CBD. These findings differed from previous studies (9,16,17) related to inflammatory conditions but were unique findings related to the application of CBD to EIMD.

Many of the previous studies that have examined CBD's effects have been conducted using animal models that examined specific medical conditions that may utilize alternative inflammatory pathways compared with EIMD (3–5,29). For example, CBD is capable of reducing neuroinflammation associated with Alzheimer's disease through glial pathways demonstrated by significantly reducing IL-1β and iNOS upregulation (4). Furthermore, inflammation associated with arthritis is largely mediated by the release of TNF-α. Malfait et al. (5) reported that, *in vivo,* when CBD injected i.p. with 10 mg·kg^{-1}, it suppressed lipopolysaccharide-induced serum TNF-α. The authors (5) suggested that CBD's immunosuppressive effects could be the result of a combination of both a T-helper 1 response and an anti-inflammatory action by way of decreasing TNF-α in the synovium. It has been demonstrated (51) that inflammation related to EIMD may be facilitated by the release of intercellular enzymes and muscle proteins such as CK, myoglobin (Mb), and lactate dehydrogenase (LDH), as well as the release of proinflammatory cytokines such as IL-1, IL-6, and TNF-α. Although CBD was able to suppress TNF-α via the A2A and CB2 receptor-mediated pathways in arthritis, other inflammatory mediators are still present, and more research is needed to understand the pathways in which CBD elicits its effects after EIMD in humans. Nonetheless, it is important to note that previous studies (4,5,29,41,53) had reported dose-dependent effects, meaning it is possible that the dosage and/or dose scheduling in the current study may not have been sufficient to produce reported therapeutic effects.

In the present study, a 150-mg dose of CBD was given to subjects via oral ingestion in two separate 75-mg doses to be separated by 8 h over the course of 4 d, for a total approximate dose of 6 mg·kg^{-1}. The dose that was administered in the present study was selected, as it is considered within the typical range of commercial supplement doses and is more commonly used on a recreational basis (57). Although 150 mg of CBD is within the typical dose range reported and has been used in previous human studies (58,59) investigating other clinical conditions, a limitation of funding prevented the use of multiple dosages and schedules within the present study. Dosages, within the present study, were also separated by 8 h. Only one previous study (17) has reported on the effect of CBD mixed with MCT oil on markers of muscle damage in trained men. This study (17) induced damage in the leg extensors and found that a significantly lower mixed dose of CBD (16 mg total mixed dose) improved perceived soreness after EIMD. However, this study did not test CBD only; thus, it is unclear how results may have differed without the confounding interaction of MCT oil. Other studies (25,60) have used similar dose schedules to investigate the effects of ibuprofen on DOMS and muscle performance after performing exercise. Nevertheless, the present study revealed that in humans a relative dose of approximately 2 mg·kg^{-1}·d^{-1} and dose schedule is likely ineffective in reducing muscle soreness, inflammation, and muscular performance in untrained college-age men. Because CBD has been shown to be dose-dependent and prescribing guidelines are lacking, more research should be done using a greater range of dosages and dose schedule for its application to EIMD.

Many studies (9,16,37) have used noninvasive markers of muscle damage to determine muscle inflammation. It is common to observe an increase in inflammatory cells after performing exercise (14,51). For example, MacIntyre et al. (14) reported an increase in the accumulation of neutrophils and cytokines, such as IL-6, after ECC exercise of the quadriceps. It

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is also common to observe an increase in intramuscular proteins and serum enzymes, such as CK, LDH, and Mb after ECC exercise. This is because the damage caused by ECC exercise causes tearing and leakage of intramuscular proteins and serum enzymes from myofibrils, which exacerbates the inflammatory response (15). Jenkins et al. (51) reported that increases in CK, LDH, and Mb observed from their study occurred as a result of muscle damage caused by the ECC exercise protocol. Studies investigating the effects of CBD typically measure the accumulation of leukocytes, and/or proinflammatory cytokines such as IL-1, IL-6, and TNF-α. However, very few have investigated CBD's effects on intramuscular proteins and enzymes, such as CK, LDH, and Mb, which occur as a result of muscle damage after exercise. Therefore, prospective studies should investigate the effects of CBD on intramuscular proteins and enzymes using more invasive measures.

In the present study, CBD supplementation at 150 $mg \cdot d^{-1}$ was found to have no effect on the noninvasive assessment of muscle soreness, inflammation, and strength performance after bouts of ECC exercise of the elbow flexors in men. Because of the unique nature of CBD oil, it is possible that the pathways in which CBD may elicit its anti-inflammatory and antihyperalgesic effects may not have any influence on the recovery of muscle function after EIMD. The lack of difference between conditions and lack of interaction between condition and time indicated that at the current dose and schedule, CBD oil may not be beneficial for untrained, college-age men as a recovery aid after EIMD. Although this study did not find a significant benefit of CBD oil, it will inform future studies investigating the efficacy of CBD oil for the relief of soreness and inflammation after exercise. It is recommended that future studies investigate wider ranges of CBD dosages and scheduling in both trained and untrained men and women. The application of invasive measurement techniques such as the collection of blood serum indices to further examine CBD's therapeutic potential as a muscle recovery aid after EIMD may also elucidate the mechanistic pathways mediated by CBD oil as well as help to differentiate perceived versus physiological effects of the supplement.

The authors declare no conflicts of interest or external funding or sponsorship related to this study. The results of the present study do not constitute an endorsement by the American College of Sports Medicine. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation.

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